

14/11/14 (E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: July 2014 - November 2014

Max. Marks: 100**Duration:****End Semester Exam**

Class: Second Year

Semester: III

Name of the Course: Database Management System

Branch: IT

Course Code: UTPC305

Instructions:

(1) All Questions are Compulsory

(2)

(3)

(4)

Question No.		Max. Marks
Q1	Explain the advantages of DBMS over a file system. OR Describe the overall System Structure Of DBMS with diagram.	10
Q2(a)	Illustrate the importance of Normalization in Database Design. OR Illustrate BCNF with an example.	10
Q2(b)	Consider relation R(PQRSTU) with following dependencies :- $P \rightarrow Q$ $ST \rightarrow PR$ $S \rightarrow U$. Apply normalization till 3NF.	10
Q3(a)	Draw and Explain Transaction State Diagram. OR Illustrate effect of Atomicity with an example.	10
Q3(b)	Show with an example that every conflict serializable schedule is view serializable but vice-versa is not true.	10

Q3(c)	Illustrate shadow paging with an example. OR Illustrate Thomas Write Rule with an example.	10
Q4(a)	Design an ER diagram for a university database consisting of 4 entities : (i) Student (iii) Class (ii) Department (iv) Faculty A student has a unique id, the student can enroll for multiple classes and has at-most one major. Faculty must belong to department & faculty can take multiple classes. Every student will get a grade for the class he/she has enrolled.	10
Q4(b)	Convert the above ER diagram in Q4(a) to tables	10
Q5(a)	For the given employee database give an expression in SQL for the following: Employee (empname, street, city) Works (empname, company-name, salary) Company (company-name, city) Managers (empname, manager-name) (i) Modify database so that 'swapnil' now lives in 'Navi Mumbai'. (ii) Give all employees of 'IBM' a 10% raise. (iii) List all the employees who live in the same cities as their managers.	10
Q5(b)	If R1 and R2 are two relations (tables) in the database, show with an example that joined relation R1 left outer join R2 produces same tuples as R2 right outer join R1.	10

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